T3 D_+2: general EFT1 RGEs

Matchete/RGBeta symbolic output in mathematical notation

D_+2

Additional quartics: 0, additional bilinears: 0, extra Yukawa: none.
General matching and Wilson coefficients are not computed in this symbolic report.

One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

g_1

$$16\pi^2 \frac{dg_1}{d\ln\mu} = \frac{55{g_1}^3}{6}$$

g_2

$$16\pi^2 \frac{dg_2}{d\ln\mu} = -\frac{5{g_2}^3}{2}$$

g_3

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

Y_u

$$\begin{aligned}
16\pi^2 \frac{dY_u}{d\ln\mu} = {} & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\
& - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\
& - 8{g_3}^2\,(Y_u)_{ij} \\
& - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}
\end{aligned}$$

Y_d

$$\begin{aligned}
16\pi^2 \frac{dY_d}{d\ln\mu} = {} & \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\
& + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\
& - 8{g_3}^2\,(Y_d)_{ij} \\
& - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}
\end{aligned}$$

Y_e

$$\begin{aligned}16\pi^2 \frac{dY_e}{d\ln\mu} = & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij} \\ & + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij} \\ & + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij} \\ & + \frac{3}{2}\Big(Y_e\ (Y_e)^\dagger\ Y_e\Big)_{ij} \\ & - \frac{9}{4}{g_2}^2\,(Y_e)_{ij} \\ & - \frac{15}{4}{g_1}^2\,(Y_e)_{ij}\end{aligned}$$

m_1^2

$$\begin{aligned}16\pi^2 \frac{dm_1^2}{d\ln\mu} = & -12{g_2}^2 m_1^2 \\ & - 6{g_1}^2 m_1^2 \\ & + 2\lambda_{12}^{(1)} m_2^2 \\ & + m_1^2(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})\end{aligned}$$

m_2^2

$$\begin{aligned}16\pi^2 \frac{dm_2^2}{d\ln\mu} = & -24{g_1}^2 m_2^2 \\ & + 6\lambda_{12}^{(1)} m_1^2 \\ & + 4\lambda_{S_2}^{(1)} m_2^2\end{aligned}$$

λ_1

$$\begin{aligned}16\pi^2 \frac{d\lambda_1}{d\ln\mu} = & \lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\ & - 12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big) \\ & - 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\ & - 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big) \\ & + 4\lambda_5\,(\lambda_5)^* \\ & + \frac{9{g_2}^4}{4} \\ & + {g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_1\right) \\ & - 9{g_2}^2\lambda_1 \\ & + \frac{3{g_1}^4}{4} \\ & + 12{\lambda_1}^2 \\ & + 6{\lambda_{HS_1}^{(1)}}^2\end{aligned}$$

$$
\begin{aligned}
&+ 3{\lambda_{HS_1}^{(A)}}^2 \\
&+ 2{\lambda_{HS_2}^{(1)}}^2
\end{aligned}
$$

$\lambda_{S_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(1)}}{d\ln\mu} = &22{g_2}^4 \\
&+ {g_1}^2\left(8{g_2}^2 - 12\lambda_{S_1}^{(1)}\right) \\
&- 24{g_2}^2\lambda_{S_1}^{(1)} \\
&+ 12{g_1}^4 \\
&+ 2{\lambda_{12}^{(1)}}^2 \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ 14{\lambda_{S_1}^{(1)}}^2 \\
&+ 8\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)} \\
&+ 16{\lambda_{S_1}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(1)}}{d\ln\mu} = &192{g_1}^4 \\
&- 48{g_1}^2\lambda_{S_2}^{(1)} \\
&+ 6{\lambda_{12}^{(1)}}^2 \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ 10{\lambda_{S_2}^{(1)}}^2
\end{aligned}
$$

$\lambda_{HS_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} = &\lambda_{HS_1}^{(1)}(6\,\mathrm{Tr}\left(Y_d\ {(Y_d)}^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ {(Y_e)}^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ {(Y_u)}^\dagger\right) + 4\lambda_{S_1}^{(A)}) \\
&+ 4\lambda_5\,{(\lambda_5)}^* \\
&+ 6{g_2}^4 \\
&- \frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 3{g_1}^4 \\
&- \frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$$+ 6{\lambda_{HS_1}^{(A)}}^2$$

$\lambda_{HS_2}^{(1)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = \lambda_{HS_2}^{(1)}(&6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))\\ &+ 12\lambda_5\,(\lambda_5)^*\\ &- \frac{9{g_2}^2\lambda_{HS_2}^{(1)}}{2}\\ &+ 12{g_1}^4\\ &- \frac{51{g_1}^2\lambda_{HS_2}^{(1)}}{2}\\ &+ 6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}\\ &+ 6\lambda_1\lambda_{HS_2}^{(1)}\\ &+ 4{\lambda_{HS_2}^{(1)}}^2\\ &+ 4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}\end{aligned}$$

$\lambda_{12}^{(1)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{12}^{(1)}}{d\ln\mu} = &4\lambda_5\,(\lambda_5)^*\\ &- 12{g_2}^2\lambda_{12}^{(1)}\\ &+ 48{g_1}^4\\ &- 30{g_1}^2\lambda_{12}^{(1)}\\ &+ 4{\lambda_{12}^{(1)}}^2\\ &+ 8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}\\ &+ 4\lambda_{12}^{(1)}\lambda_{S_1}^{(A)}\\ &+ 4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}\\ &+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}\end{aligned}$$

λ_5

$$\begin{aligned}16\pi^2 \frac{d\lambda_5}{d\ln\mu} = \lambda_5(&6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) - 4\lambda_{HS_1}^{(A)})\\ &- \frac{21{g_2}^2\lambda_5}{2}\\ &- \frac{33{g_1}^2\lambda_5}{2}\\ &+ 2\lambda_{12}^{(1)}\lambda_5\\ &+ 2\lambda_1\lambda_5\\ &+ 4\lambda_{HS_1}^{(1)}\lambda_5\\ &+ 4\lambda_{HS_2}^{(1)}\lambda_5\end{aligned}$$

$\lambda_{HS_1}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(A)}}{d\ln\mu} = \lambda_{HS_1}^{(A)}(&6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 16\lambda_{S_1}^{(A)}) \\
&- \frac{8}{3}\lambda_5\,(\lambda_5)^* \\
&+ {g_1}^2\left(4{g_2}^2 - \frac{15\lambda_{HS_1}^{(A)}}{2}\right) \\
&- \frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2} \\
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$\lambda_{S_1}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(A)}}{d\ln\mu} = &-2{g_2}^4 \\
&+ {g_1}^2\left(8{g_2}^2 - 12\lambda_{S_1}^{(A)}\right) \\
&- 24{g_2}^2\lambda_{S_1}^{(A)} \\
&+ 3{\lambda_{HS_1}^{(A)}}^2 \\
&+ 12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)} \\
&+ 20{\lambda_{S_1}^{(A)}}^2
\end{aligned}
$$